

6. Computer Ethics

Data Privacy and Responsibilities

Introduction

Computer ethics means the moral responsibility connected with the use, design, and control of computer systems. Today computers are used in banks, hospitals, universities, industries, transport, and defense systems, so misuse of computers can harm many people.

According to Fleddermann, computer ethics problems are mainly of two types: computers used to commit unethical acts, and computers used improperly as engineering tools. Therefore, computer engineers must think about privacy, security, honesty, safety, and professional judgment.

1. Computers as Tools for Unethical Acts

Computers can make unethical actions easier, faster, and harder to trace. For example, stealing money through a banking system is ethically the same as physical theft, but computers can make the crime larger and more hidden.

The main point is that computers do not change the moral issue. Theft, fraud, deception, hacking, and misuse of information are still wrong, even if they are done through software or the internet.

Common unethical acts include:

- hacking into systems
 - stealing money or data
 - spreading viruses
 - changing records
 - misusing private information
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2. Data Privacy

Data privacy means a person's right to control access to information about themselves. This includes medical records, financial data, passwords, academic records, location, personal messages, and identity details.

Fleddermann explains that computers make privacy harder to protect because large amounts of personal and corporate data are stored in central systems, where many people may access them. Computerization makes data easier to use for valid purposes, but also easier to misuse.

3. Why Privacy Is an Ethical Issue

Privacy is ethical because private information can harm a person if it is leaked or misused. A person may face embarrassment, harassment, blackmail, identity theft, or unfair treatment.

Fleddermann gives two main reasons privacy matters. First, leaking private information can lead to harassment or blackmail. Second, personal information can be treated like personal property, so unauthorized use of it is similar to theft.

4. Responsibilities of Computer Engineers

Computer engineers must protect the data and systems under their control. They should design systems that prevent unauthorized access, leakage, corruption, and misuse of information.

Their responsibilities include:

- collect only necessary data
- give access only to authorized people
- use passwords, authentication, and encryption
- keep backups and logs
- report security risks honestly
- protect corporate and user confidentiality

For example, a university system should not allow every staff member to view students' private records.

5. Hacking and Unauthorized Access

Hacking means entering or using a computer system without permission. Even if no visible damage is done, hacking is unethical because it violates privacy, ownership, and security.

Fleddermann explains that hacking includes unauthorized access to databases, changing information, adding false information, and spreading viruses. Even when hackers are only "testing their skill," hacking is still an unethical use of computers.

6. Copyright and Intellectual Property

Computer ethics also includes respect for software ownership and intellectual property. Software code, algorithms, designs, and technical documents may belong to an employer, client, or company.

Using copied code, pirated software, or a previous employer's protected work without permission is unethical. Copyright exists to protect creators and their right to benefit from their work.

7. Responsibility When Using Software Tools

Engineers often use software for design, simulation, testing, and analysis. However, software should not replace engineering judgment.

Fleddermann warns that software can give an engineer the false feeling that they are qualified in an area beyond their expertise. An engineer must understand the limits of software and must verify results before trusting them.

8. Safety-Critical and Autonomous Systems

Many systems now depend on embedded computers, such as cars, medical machines, aircraft, and industrial control systems. If these systems fail, people may be injured or killed.

Fleddermann says engineers must properly test software, allow human intervention when needed, and provide hardware backup instead of depending only on software. In autonomous systems, human control is important because computers making decisions alone can create serious harm.

Conclusion

Computer ethics is important because computer systems store private data, control important services, and affect public safety. A computer engineer should not think only about technical performance.

A responsible engineer must protect privacy, avoid hacking, respect copyright, test software properly, verify results, and keep human control in critical systems. In this way, computer systems can be useful, safe, and ethically responsible.

How to Apply in Case Studies

In a computer ethics case, first identify the main issue: privacy, hacking, copyright, software failure, data misuse, or unsafe automation.

Then identify who is affected: users, patients, customers, students, employer, public, or government.

After that, apply the duty. If data is leaked, write about privacy and confidentiality. If code is copied, write about copyright. If software fails, write about testing, safety, and engineering responsibility.

Final answer should clearly say that the engineer must protect privacy, respect ownership, use proper security, test software, and protect the public.

Short Case Study: Hospital Data Privacy

A hospital software engineer has access to patient records and shares a patient's medical history with a friend.

This is unethical because medical data is private information. The patient has the right to control access to personal information. Sharing it without permission violates privacy and confidentiality, and may cause embarrassment or harm.

The correct action is to keep patient data confidential and share it only with authorized medical staff for proper medical reasons.

Memorization Checklist

Remember computer ethics with:

P-H-C-S-A

P — Privacy: Protect personal and corporate data.
H — Hacking: Avoid unauthorized access.
C — Copyright: Respect software and intellectual property.
S — Software safety: Test software and verify results.
A — Autonomy: Keep human control in critical systems.

5-Mark Exam Answer

Computer ethics means the moral responsibility connected with the use, design, and control of computer systems. According to Fleddermann, computer ethics problems are mainly of two types: computers used to commit unethical acts and computers used improperly as engineering tools. Computers can make unethical acts like theft, fraud, hacking, and data misuse easier and harder to trace.

Data privacy means a person's right to control access to information about themselves. Computers store large amounts of personal and corporate data, so engineers must protect this information from unauthorized access. Privacy invasion can cause harassment, blackmail, identity theft, and misuse of personal property.

Computer engineers must avoid hacking, protect confidential data, respect copyright and intellectual property, and use software tools responsibly. They must test software properly, verify computer results, and provide human control and backup systems in safety-critical applications. Therefore, a responsible computer engineer protects privacy, security, honesty, public safety, and professional judgment.